

LESSON 5-5 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2**Volume of Rectangular Prisms**

Maryland Standard 6.G.A.2 · Learning target – I can find the volume of a rectangular prism by multiplying length, width, and height.

REMEMBER

- **Volume** counts the unit cubes that fill a solid. Use **cubic units** (cm^3 , in^3 , ft^3).
- **$V = l \times w \times h$** , or **$V = B \times h$** , where B is the area of the base.

WORKED EXAMPLE

A box is 7 cm long, 3 cm wide, and 5 cm tall.

Step 1 One layer of cubes: $B = 7 \times 3 = 21$

Step 2 Stack 5 layers: $21 \times 5 = 105$

Answer 105 cm^3

WORD BANK

volume – the space inside a solid

cubic units – cm^3 , in^3 , ft^3

base area – length $\times$ width of the bottom

layer – one flat level of cubes

PRACTICE

show your work · label units

1. Each small cube is 1 cm^3 . Find the volume of the prism.

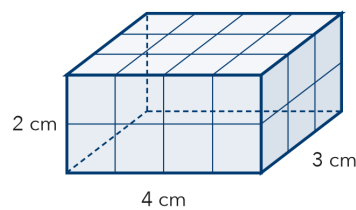
Step 1 Count the cubes in one layer: 4×3 .

Step 2 Count the layers.

Step 3 Multiply.

Show your work.

Answer _____ cm^3



2. Find the volume of each rectangular prism.

a. $8 \text{ in} \times 5 \text{ in} \times 2 \text{ in}$ _____ in^3

b. $10 \text{ cm} \times 10 \text{ cm} \times 4 \text{ cm}$ _____ cm^3

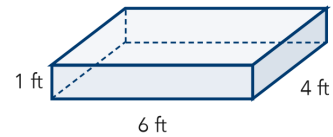
c. $6 \text{ ft} \times 6 \text{ ft} \times 6 \text{ ft}$ _____ ft^3

d. $B = 45 \text{ m}^2$, $h = 7 \text{ m}$ _____ m^3

e. $12 \text{ m} \times 3 \text{ m} \times 5 \text{ m}$ _____ m^3

f. $B = 32 \text{ in}^2$, $h = 9 \text{ in}$ _____ in^3

3. **Sandbox.** A sandbox is 6 ft long, 4 ft wide, and 1 ft deep. One bag of sand fills 2 ft^3 . How many **bags** fill the sandbox?



Step 1 Find the volume: $6 \times 4 \times 1$.

Step 2 Divide by 2 ft^3 for each bag.

Show your work.

Answer _____ bags

4. **Fish Tank.** A fish tank is 20 in long and 10 in wide. It holds $3,000 \text{ in}^3$ of water when full. How tall is the tank?

Step 1 Find the base: 20×10 .

Step 2 Divide 3,000 by the base.

Answer _____ in

5. **Compare.** Box A is 6 in $\times$ 4 in $\times$ 2 in. Box B is 3 in $\times$ 4 in $\times$ 4 in. Mika says Box A holds more because it is longer. Is she right? Explain.

Frame Mika is ____ because Box A holds ____ in^3 and Box B holds ____ in^3 .

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____

AT HOME Find a box at home. Measure or estimate its length, width, and height, and find its volume.

LESSON 5-5 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2**Missing Fractional Edges**

Maryland Standard 6.G.A.2 · Learning target – I can find a missing edge of a prism and find volumes with fractional edge lengths.

REMEMBER

- **$V = l \times w \times h$** works with fractions too. Write mixed numbers as fractions first.
- Missing edge: **divide** the volume by the product of the other two edges.

WORKED EXAMPLE

A box is $2\frac{1}{2}$ ft $\times$ 2 ft $\times$ $1\frac{1}{2}$ ft. Find its volume.

Step 1 Write as fractions: $\frac{5}{2} \times 2 \times \frac{3}{2}$

Step 2 Multiply: $(\frac{5}{2} \times 2 \times \frac{3}{2}) / (2 \times 1 \times 2) = \frac{30}{4}$

Step 3 Simplify: $\frac{30}{4} = 7\frac{1}{2}$

Answer $7\frac{1}{2}$ ft³

WORD BANK

fraction edge – an edge like $\frac{1}{2}$ ft or $1\frac{1}{2}$ in

missing edge – the edge you do not know

divide – $\div$ undoes $\times$

product – the answer when you multiply

PRACTICE

show your work · label units

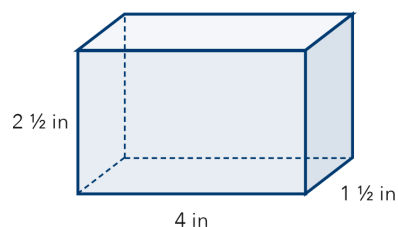
1. Find the volume of the prism.

Step 1 Write $1\frac{1}{2}$ as $\frac{3}{2}$ and $2\frac{1}{2}$ as $\frac{5}{2}$.

Step 2 Multiply $4 \times \frac{3}{2} \times \frac{5}{2}$.

Show your work.

Answer _____ in³



2. Find each volume.

a. $\frac{1}{2}$ ft $\times$ 6 ft $\times$ 8 ft _____ ft³

b. $3\frac{1}{2}$ m $\times$ 2 m $\times$ 4 m _____ m³

c. $2\frac{1}{2}$ cm $\times$ 4 cm $\times$ 3 cm _____ cm³

d. $1\frac{1}{2}$ in $\times$ $1\frac{1}{2}$ in $\times$ 4 in _____ in³

e. 4 ft $\times$ $\frac{1}{2}$ ft $\times$ 5 ft _____ ft³

f. 6 cm $\times$ $1\frac{1}{2}$ cm $\times$ 3 cm _____ cm³

3. **Planter.** A planter box is 3 ft long and $1\frac{1}{2}$ ft wide. It holds 9 ft³ of soil. How **deep** is it?

Step 1 Multiply the two edges: $3 \times 1\frac{1}{2}$.

Step 2 Divide 9 by that answer.

Answer _____ ft

- 4. Small Cubes.** A box is 2 in long, 1 in wide, and $1\frac{1}{2}$ in tall. It is packed with cubes that are $\frac{1}{2}$ in on each edge.

Step 1 Count the cubes along each edge: $2 \div \frac{1}{2}$, $1 \div \frac{1}{2}$, $1\frac{1}{2} \div \frac{1}{2}$.

Step 2 Multiply to get the number of cubes.

- a. How many cubes fill the box? _____
- b. Each cube is $\frac{1}{8}$ in³. What is the volume of the box? _____ in³

TEST PRACTICE

choose the best answer

- 5.** A box is 5 in long, 2 in wide, and $1\frac{1}{2}$ in tall. What is its volume?

A $8\frac{1}{2}$ in³ **B** 10 in³ **C** 15 in² **D** 15 in³

- 6.** A prism has a volume of 96 ft³. Its base is 4 ft by 6 ft. What is its height?

Step 1 Find the base: 4×6 .

Step 2 Divide 96 by the base.

A 4 ft **B** 16 ft **C** 24 ft **D** 576 ft

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____

AT HOME Find a box at home with a fraction edge (like $2\frac{1}{2}$ in). Estimate its volume.